

The ER diagram represents the core entities involved in the Loan Approval Prediction System and illustrates how they interact with each other. It provides a clear structure for organizing applicant information, loan requests, prediction results, and machine learning model outputs within the database.

Entities Involved

The ER diagram features seven primary entities:

User

SoilData

Crop

Dataset

MLModel

Prediction

Report

Primary Keys

Each entity is uniquely identified by its primary key:

User: user_id

SoilData: soil_id

Crop: crop_id

Dataset: dataset_id

MLModel: model_id

Prediction: prediction_id

Report: report_id

Relationships

User to SoilData: One user can submit multiple soil data records for crop analysis (1 to Many).

SoilData to Prediction: A single soil data entry can generate one crop prediction result (1 to One).

Crop to Prediction: One crop can be recommended in multiple prediction results (1 to Many).

MLModel to Prediction: One machine learning model can generate multiple prediction records (1 to Many).

Dataset to MLModel: A single dataset can be used to train multiple machine learning models (1 to Many).

Prediction to Report: One prediction can generate multiple agricultural reports and recommendations (1 to Many).

Foreign Keys

SoilData references User via user_id.

Prediction references SoilData via soil_id.

Prediction references Crop via crop_id.

Prediction references MLModel via model_id.

Report references Prediction via predictio_id.

Cardinality

A single user can submit multiple soil data records for agricultural analysis. Each soil data entry is processed by the machine learning model to generate crop prediction results. A crop can appear in multiple predictions, while a single machine learning model can generate predictions for many users and soil conditions. Each prediction can further generate multiple agricultural reports containing summaries and farming recommendations.

Normalization and Structure

The ER diagram is normalized to separate user information, soil parameters, crop details, datasets, machine learning models, prediction records, and reports into different entities. This structure reduces redundancy, improves scalability, and ensures efficient management of agricultural and prediction-related data.

Use Case Coverage

This ER model supports the core functionalities of the OptiCrop Smart Agricultural Production Optimization System, including:

Collecting and managing soil and environmental data from users.

Predicting suitable crops using machine learning algorithms.

Managing datasets and trained machine learning models.

Generating intelligent crop recommendations and prediction reports.

Supporting sustainable farming practices and data-driven agricultural decision-making.